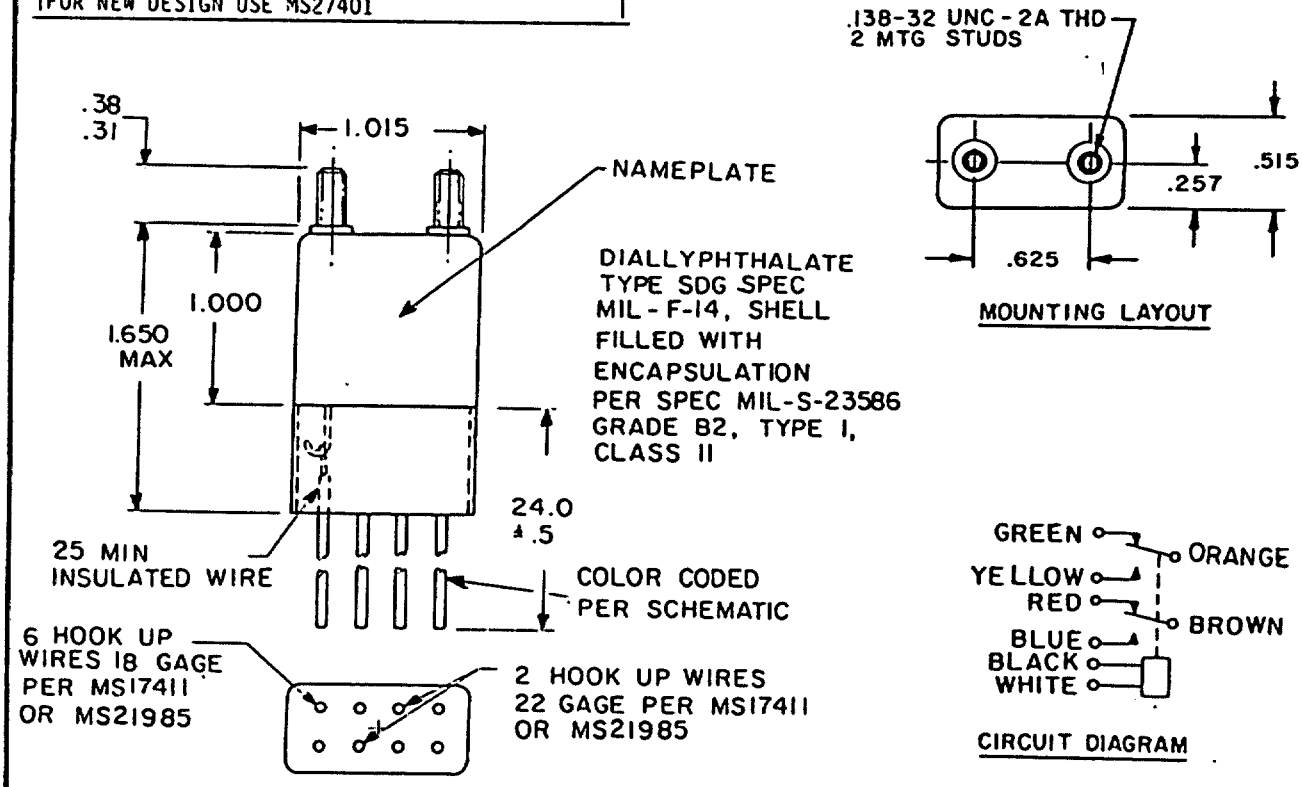
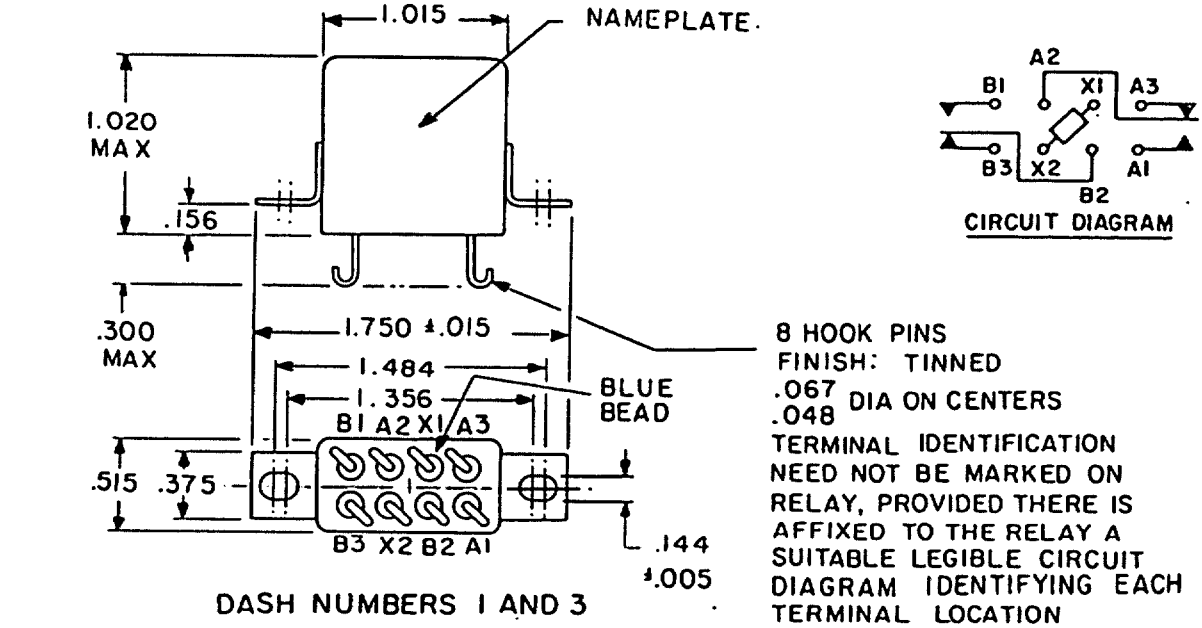
INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD
FOR NEW DESIGN USE MS27401

FED. SUP CLASS
5945




DASH NUMBERS 2 AND 4




DASH NUMBERS 1 AND 3

Ⓜ denotes changes

| | | | |
|---|---|---|---|
| P.A USAF - 85<br>Other Cust<br>Navy - EC | International interest | TITLE<br>RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED | MILITARY STANDARD<br>MS27245 |
| Procurement Specification<br>MIL-R-6106 | | SUPERSEDES: | PAGE 1 OF 8 |

DD FORM 1 MAY 73 672 (Coordinated previous editions of this form are obsolete)
*SC N/A

5945-0798-01

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

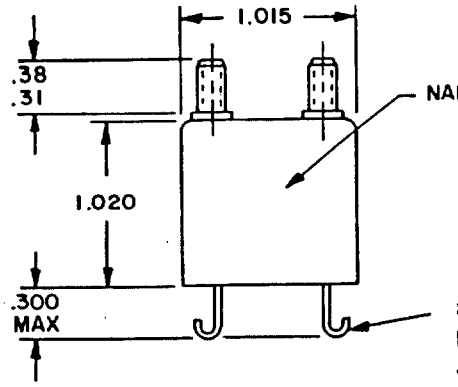
APPROVED 10 Aug 1964 REVISED Ⓛ 5 JUN 87 Ⓜ 10 FEB 89

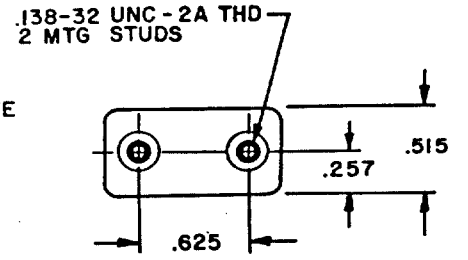
User activities: Army - Navy - Air Force -

Review activities: Army - AS Navy - Air Force - 11, 99

Military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications for repetitive use shall be made from this document when applicable.

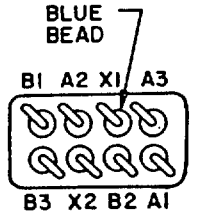
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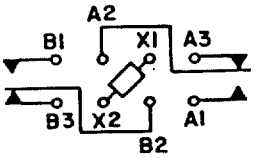



MOUNTING LAYOUT

8 HOOK PINS
FINISH: TINNED
.067
.048 DIA ON .200 CENTERS
TERMINAL IDENTIFICATION NEED NOT BE MARKED ON RELAY, PROVIDED THERE IS AFFIXED TO THE RELAY A SUITABLE LEGIBLE CIRCUIT DIAGRAM IDENTIFYING EACH TERMINAL LOCATION




CIRCUIT DIAGRAM

## DASH NUMBERS 5 AND 6

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .005 | 0.13 | .156 | 3.96 | .38 | 9.7 | 1.020 | 25.91 |
| .015 | 0.38 | .25 | 6.4 | .5 | 13. | 1.356 | 34.44 |
| .048 | 1.22 | .257 | 6.53 | .515 | 13.08 | 1.484 | 37.69 |
| .067 | 1.70 | .300 | 7.62 | .625 | 15.88 | 1.650 | 41.91 |
| .138 | 3.51 | .31 | 7.9 | 1.000 | 25.40 | 1.750 | 44.45 |
| .144 | 3.66 | .375 | 9.52 | 1.015 | 25.78 | 24.0 | 610. |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
5. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

APPROVED 10 Aug 1964
REVISED (M) For changes see pages 3,5,6 and 8

User activities: Army - Navy - Air Force -
Review activities: Army - AS Navy - Air Force - 11, 99

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

| P.A USAF - 85 Other Cust Navy - EC | International interest | TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED | MILITARY STANDARD MS27245 |
|---|---|---|---|
| Procurement Specification MIL-R-6106 | | SUPERSEDES: | PAGE 2 OF 8 |



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Review activities: Army -
Navy - AS
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -



TABLE I. Dash numbers and characteristics.

| Dash number MS25245- | Type | Coil | Terminal type | Mounting socket | Auxiliary contacts | Max weight in pounds |
|---|---|---|---|---|---|---|
| 1 | I | dc | Solder hook | Flange | N/A | 0.08 |
| 2 | I | dc | Wire lead | Stud | N/A | 0.23 |
| 3 | IER | dc | Solder hook | Flange | N/A | 0.08 |
| 4 | I | dc | Wire lead | Stud | N/A | 0.23 |
| 5 | I | dc | Solder hook | Stud | N/A | 0.23 |
| 6 | I | dc | Solder hook | Stud | N/A | 0.23 |

(M) TABLE II. Operating characteristics (-1 and -3).

| MS part no. | Coil data | | | | | | | | | | | Time (milliseconds-maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop-out voltage 2/ | Operate 3/ | Release 4/ | Contact bounce | | | |
| | | Volts 1/ | Freq. Hz | Res Ω ±10% | Volts | Amperes | Normal 2/ | High temp test | Cont current test | | | | | Main | | Aux | |
| | | | | | | | | | | | | | | NO | NC | NO | NC |
| MS27245-1 | X1, X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | | |
| MS27245-3 | X1, X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | | |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.



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Review activities: Army -
Navy - AS
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Procurement Specification MIL-R-6106

P.A USAF - 85
Other Cust
Navy - EC

International interest

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, IO AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD
MS27245

PAGE 4 OF 8

TABLE III. Rated contact load (amperes per pole) (case grounded) (-1 and -3)

| Type of load | Life operating cycles X $10^3$ | 28 V dc | | | | 115 V ac, I phase | | | | 115/200 V ac, 3 phase 1/ (Not applicable to -1) | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Inductive | 100 | 4 | 4 | | | 4 | | | | 4 | | | | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 100 | 3 | 3 | | | 3 | | | | 3 | | | | |
| Lamp | 100 | 2 | 2 | | | 2 | | | | 2 | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 2.5 | 2.5 | | | | | | | | | | | |
| Intmd current | 50 | | | | | | | | | | | | | 3/ |

1/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.
3/ Minimum current ratings 0.1 ampere resistive, 0.3 ampere inductive, and 0.5 ampere resistive are applicable in accordance with acquisition specifications except monitored to a 3 ohm contact resistance.

FED. SUP CLASS 5945

APPROVED 10 Aug 1964 REVISED (M) For changes see pages 3,5,6 and 8

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Review activities: Army - Navy - AS Air Force - 11, 99

User activities: Army - Navy - Air Force -

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Procurement Specification MIL-R-6106

P.A USAF - 85 Other Cust Navy - EC

International Interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD MS27245

PAGE 5 OF 8

FED. SUP CLASS 5945

## Environmental characteristics

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 G |
| Duration | 11 ms |
| Max duration contact opening | 10 µs |
| Vibration - sinusoidal | |
| G-level | 20 G |
| Frequency range | 10 - 2,000 Hz |
| Acceleration | 15 G |

## Electrical characteristics

| | |
|---|---|
| Insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level)

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,250 V rms | 1,000 V rms |

Dielectric strength (altitude)

| | 80,000 ft |
|---|---|
| Coil to case | 350 V rms |
| Aux contacts | |
| All other points | 350 V rms |

| | |
|---|---|
| Max contact drop initial | 0.150 volt |
| After life test | 0.175 volt |
| Overload current | 20 amperes, 28 V dc |
| Rupture current | 40 amperes, 115 V ac, 400 Hz |
| Duty rating | Continuous |

## Quality conformance inspection

Performance of groups B and C tests are not applicable.

(M) Group A acceptance reports shall be submitted to the qualifying activity on a yearly basis in order to retain qualification for this military standard sheet.

## DASH NUMBERS 1 AND 3

APPROVED 10 Aug 1964 REVISED (M) For changes see pages 3,5,6 and 8

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Review activities: Army -
Navy - AS
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -



(M) TABLE IV. Operating characteristics (-2. -4, -5, and -6).

| MS part no. | Coil data | | | | | | | | | | | Time (milliseconds-maximum)2/ | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop-out voltage 2/ | Operate 3/ | Release 4/ | Contact bounce | | | |
| | | Volts 1/ | Freq. Hz | Res Ω ±10% | Volts | Amperes | Normal 2/ | High temp test | Cont current test | | | | | Main | | Aux | |
| | | | | | | | | | | | | | | NO | NC | NO | NC |
| MS27245-2 | X1, X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | | |
| MS27245-4 | X1, X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | | |
| MS27245-5 | X1, X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | | |
| MS27245-6 | X1, X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | | |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.



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Review activities: Army -
Navy - AS
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

TABLE V. Rated contact load (amperes per pole) (case grounded) (-2, -4, -5, and -6).

| Type of load | Life operating cycles X $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ (not applicable to -2 and -6) | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 50 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Inductive | 50 | 6 | 6 | | | 6 | | | | 4 | | | | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 50 | 4 | 4 | | | 4 | | | | 3 | | | | |
| Lamp | 50 | 2 | 2 | | | 2 | | | | 2 | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 2.5 | 2.5 | | | | | | | | | | | |
| Intmd current | 50 | | | | | | | | | | | | | 3/ |

1/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.
3/ Minimum current ratings 0.1 ampere resistive, 0.3 ampere inductive, and 0.5 ampere resistive are applicable in accordance with acquisition specifications except monitored to a 3 ohm contact resistance.

DD FORM 672 1 MAY 73 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

P.A USAF - 85
Other Cust Navy - EC

International interest

Procurement Specification MIL-R-6106

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD
MS27245

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FED. SUP CLASS 5945

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Review activities: Army -
Navy - AS
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

DD FORM 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 1 MAY 73

Procurement Specification MIL-R-6106

P.A USAF - 85
Other Cust Navy - EC

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD

MS27245

PAGE 8 OF 8

## Environmental characteristics

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 G |
| Duration | 11 ms |
| Max duration contact opening | 10 µs |
| Vibration - sinusoidal | |
| G-level | 20 G |
| Frequency range | 10 - 2,000 Hz |
| Acceleration | 15 G |

## Electrical characteristics

| | |
|---|---|
| Insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level)

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,250 V rms | 1,000 V rms |

Dielectric strength (altitude)

| | 80,000 ft |
|---|---|
| Coil to case | 350 V rms |
| Aux contacts | |
| All other points | 350 V rms |

| | |
|---|---|
| Max contact drop initial | 0.350 volt |
| After life test | 0.375 volt |
| Overload current | 20 amperes, 28 V dc |
| Rupture current | 40 amperes, 115 V ac, 400 Hz |
| Duty rating | Continuous |

## Quality conformance inspection

Performance of groups B and C tests are not applicable.

(M) Group A acceptance reports shall be submitted to the qualifing act on a yearly basis in order to retain qualification for this milita standard sheet.

## DASH NUMBERS 2, 4, 5, AND 6

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